



MODULE CODE: CSN2264

MODULE TITLE: COMPUTER NETWORKS

CAT2:

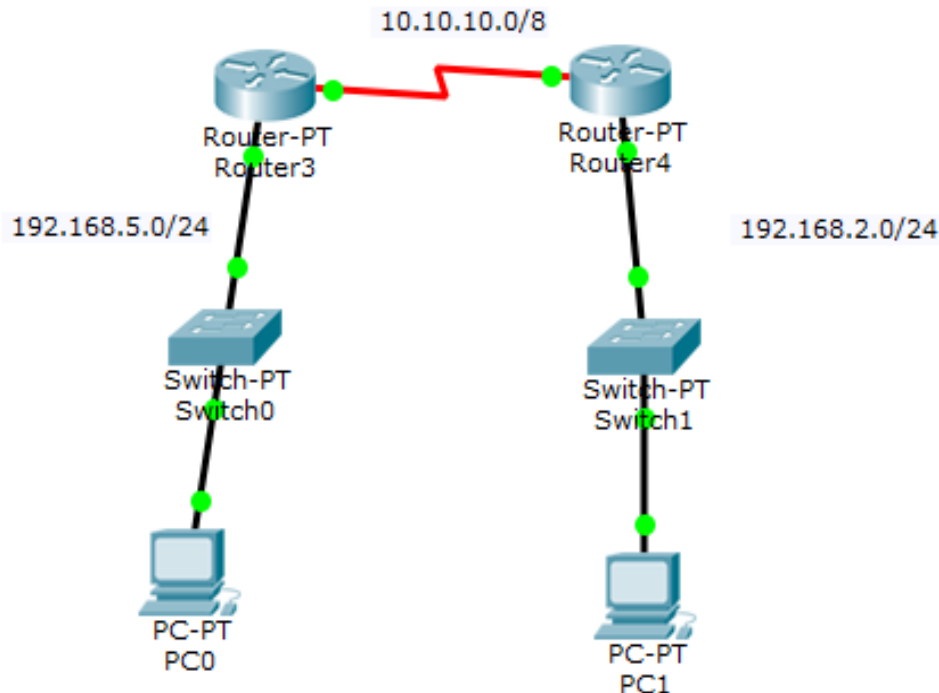
DATE: 21/11/2024

Question 01 (20 marks)

- (a) Differentiate client/server model from peer-to-peer network model. **(2 marks)**
- (b) Define a protocol and explain three elements of a protocol. **(4 marks)**
- (c) What is the function of SMTP & POP3 protocols? **(2 marks)?**
- (d) Differentiate distance vector routing protocols from link-state routing protocols. **(2 marks)**
- (e) Briefly describe two error correction methods. **(4 marks)**
- (f) Explain the difference between Transport and network layers. **(2 marks)**
- (g) Briefly differentiate two transport layer protocols. **(4 marks)**

Question 02 (15 marks)

- (a) Provide three main function of data link layer. **(3 marks)**
- (b) Provide one advantage and one dis-advantage of Dynamic Routing. **(2 marks)**
- (c) Consider the network scenario below:



- i) How many networks are found in this scenario? **(1 mark)**
- ii) Specify each network with its network ID. **(3 marks)**
- iii) Write the commands to configure RIP protocol on this scenario to allow the communication between PC0 and PC1. **(6 marks)**

Question 03 (15 marks)

- (a) With help of diagrams, describe three types of guided transmission media. **(4 marks)**
- (b) Give the differences between TCP and UDP transport layer protocols. **(3 marks)**
- (c) What is the difference in terms of functionality between the upper layers and lower layers of the OSI model? **(3 marks)**
- (d) With the help of diagrams, explain the types of errors. **(5 marks)**

Question 04 (15 marks)

- (a) What is the Network ID of the network 172.20.243.138/27 belongs to? **(2 marks)**
- (b) What is the broadcast address of the network 172.26.103.32 255.255.255.224? **(2 marks)**
- (c) What is the Subnet Mask corresponding to a shorthand of /11? **(2 marks)**

(d) Given IP address: 192.168.1.15/26

(i) Compute the subnet mask for this IP.

(ii) What are the total hosts?

(2 marks)

(iii) How many are valid?

(1 mark)

(iv) What is the Broadcast ID?

(2 marks)

(v) What are the First and Last Usable IPs?

(2 marks)

(e) Write the IP address 135.1.1.25 mask 255.255.248.0 in CIDR notation.

(2 marks)